

The role of the parahippocampal place area in processing static versus navigational scenes in a naturalistic movie

Rachel Yoon

Oxford Academy, CA

ABSTRACT

The parahippocampal place area (PPA) has been studied intensively for its role in scene perception, yet its broader contribution to the visual network remains unclear. Previous studies have highlighted the PPA's unique anatomical location in-between the ventral and dorsal streams—two highly interactive visual pathways. This study investigates whether the PPA, in alignment with its anatomical position, coordinates information flow between the two streams, acting as an “integration hub” for scene perception. Here, using naturalistic movie-watching data from the StudyForrest dataset, I analyze the PPA's functional connectivity with both scene- and object-selective regions under two distinct conditions: static scenes with limited movement versus navigational scenes with high dynamicity. The results of the experiment support the hypothesis that the PPA serves as a crucial connector of information flow between the ventral and dorsal pathways, with the right hemispheric PPA playing a larger integrative role compared to the left hemispheric PPA. Furthermore, the viewing of static scenes evoked stronger functional connectivity across the entire network, suggesting a more globally integrated brain response during static visual processing. In contrast, navigational scenes unexpectedly displayed weaker overall functional connectivity but a higher modular network organization, indicating that brain regions prioritize specific connections to maximize speed and efficiency during dynamic visual processing. These results demonstrate that the PPA functions as an integration hub within the visual network, modulating functional connectivity based on scene dynamics to facilitate adaptive processing between the ventral and dorsal pathways.

Introduction

At first glance, the human brain streamlines a vast array of visual information through a complex neural network. One category of visual information—scenes—is processed by a unique group of “scene-selective” brain areas, crucial for understanding our surrounding environment. Notably, the parahippocampal place area (PPA), a subregion of the parahippocampal cortex (PHC), is widely regarded for its prominent role in scene recognition.

The idea of a “scene-selective” region in the brain was first introduced by Epstein and Kanwisher (1998). After presenting participants with a variety of visual stimuli and examining their brain activity in response to each, the study revealed that the parahippocampal cortex—and the PPA in particular—showed significantly greater activation for images of places and scenes, rather than objects or faces (Epstein & Kanwisher, 1998; Aguirre et al., 1998; Levy et al., 2001). This selective response indicated that regions such as the PPA play a more specialized function beyond general visual processing, sparking a line of research dedicated to further defining this role.

Following Epstein and Kanwisher (1998), however, subsequent studies produced nuanced interpretations of the PPA's scene-selectivity. One study, Bar et al. (2008), compared scenes with clear contextual associations, such as a floaty in a pool or cart at a grocery store, to scenes with weak contextual associations, such as a random object placed in front of a wall. The study found that the PPA activates preferentially for high-context scenes rather than low-context scenes. Considering that place information is equated across both conditions, Bar and colleagues proposed that PPA activation is not driven by the presence of places, but rather by the strength of the scene's contextual association (Bar & Aminoff, 2003; Bar, 2004; Aminoff et al., 2007; Bar et al., 2008).

However, this interpretation was later challenged by Mullally and Maguire (2011), who proposed that the parahippocampal cortex (PHC), in its entirety, plays a more basic role in scene-selectivity. The study categorized objects as either “space-defining,” evoking a clear sense of surrounding space (i.e. walls or large furniture), or “space-ambiguous,” lacking cues that define a specific background (i.e. small, portable items). When prompting participants to imagine both types of objects in an isolated state, the researchers found that space-defining objects engaged the PHC far more, suggesting that scene processing is dependent on the basic awareness of 3D space, not

on complex properties like scene category or contextual association (Mullally & Maguire, 2011; Kravitz, Peng, et al., 2011).

Other studies have further complicated our understanding of the PPA. For example, Kravitz, Peng, et al. (2011) examined how the PPA responds to different scenic properties, such as expanse (open or closed space), distance (near or far), and content (manmade or natural). They found that the PPA's response is modulated primarily by expanse, suggesting that it reflects the spatial properties of a scene (Epstein & Kanwisher, 1998; Epstein et al., 1999; Epstein & Ward, 2010; Park & Chun, 2009; Park et al., 2011; Kravitz, Peng, et al., 2011). Following this discovery, Kamps et al. (2016) demonstrated that responses from the PPA are stronger for intact, coherent spatial layouts rather than rearranged or fragmented spaces, thus narrowing down its sensitivity to global spatial properties (i.e. the overall layout and structure of a scene) over the local details (i.e. individual elements and objects that make up the scene) (Epstein & Kanwisher, 1998; Epstein et al., 1999; Kravitz, Peng, et al., 2011; Park et al., 2011). These studies, altogether encompassing over a decade of research, strongly suggest that the PPA is a key player in scene recognition and perception, though its specific contribution remains unclear.

Since these earlier findings, research on the PPA has evolved beyond just its functional role. A newer hypothesis proposed by researchers is that the PPA serves as more of an “integration hub” rather than a bluntly scene-selective region (Harel et al., 2013; Baldassano et al., 2013). This idea stems from the PPA's unique anatomical location in-between two distinct, yet highly interactive neural pathways that aid in visual processing: the ventral (“what”) pathway, which is central to object identification, and the dorsal (“where”) pathway, which processes spatial and navigational information (Ungerleider & Mishki, 1982; Goodale & Milner, 1992; Haxby et al., 1994; Milner, 2017). Anatomically, the PPA is placed where the ventral and dorsal pathways are thought to converge, and functionally, it may serve as a “bridge” between the two networks (Ungerleider et al., 2008; Kravitz, Peng, et al., 2011; Harel et al., 2013).

One natural question is, then, which other brain regions communicate with this “hub” to establish the representation of a scene? Previous work has identified a few other brain regions in the ventral and dorsal pathways that are also important for scene perception. The occipital place area (OPA), for instance, lies along the dorsal pathway and is known for its specialization in “local” scene processing (Nakamura et al., 2000; Hasson et al., 2003; Dilks et al., 2013; Kamps et al., 2016). Kamps et al. (2016) found that the OPA represented local, scene-related elements of the spatial boundary (i.e. walls, floor, furniture), regardless of their arrangement. Unlike the PPA, the OPA is attuned to the detailed elements of a scene's immediate surroundings, making it a key player in local navigation (Kamps et al., 2016; Julian et al., 2016). Another significant region along the dorsal pathway is the retrosplenial cortex (RSC), also characterized as a strictly scene-selective region (Vogt & Pandya, 1987; Kobayashi & Amaral, 2003, 2007; Dilks et al., 2011; Kravitz, Saleem, et al., 2011; Kravitz, Peng, et al., 2011). Studies revealed the RSC to have similar tendencies as the PPA in terms of its heightened response to global spatial properties, except it plays a greater role in spatial memory and navigation (Ino et al., 2002), particularly in integrating the scene layout into broader spatial contexts (Maguire, 2001; Byrne et al., 2007; Epstein, Parker, et al., 2007; Epstein, 2008; Vann et al., 2009; Dilks et al., 2011; Kamps et al., 2016).

In addition to the RSC and OPA, the lateral occipital complex (LOC), positioned inside the ventral pathway, traditionally considered to be a region for intermediate-level object perception (Malach et al., 1995; Kim et al., 2009; Pitcher et al., 2009; Margalit, 2016), has gained attraction in scene perception research. This is because recent studies have demonstrated a mutually beneficial relationship between object and scene processing, indicating that there exists some form of communication between the ventral and dorsal pathways (Troiani et al., 2014; Brandman & Peelen, 2017; Aminoff & Durham, 2022; Brandman & Peelen, 2023; Peelen et al., 2024). An MEG study, for instance, found that blurry scenes become clearer when paired with objects (Brandman & Peelen, 2023). On the other hand, scenes were found to aid in object recognition by providing contextual association that catalyzes visual processing (Biederman et al., 1982; Bar, 2004; Oliva & Torralba, 2007; Smith & Federmeier, 2024; Peelen et al., 2024). Studies have also found that object and scene information are not mutually exclusive with one another; for instance, Harel et al. (2013) interestingly found that the PPA was sensitive to the presence or absence of both objects and spatial backgrounds, which complicates an earlier view of the PPA as strictly “scene-selective” (Epstein & Kanwisher, 1998; Aguirre et al., 1998; Levy et al., 2001; Kravitz, Peng, et al., 2011; Park et al., 2011).

A study by Watson and Andrews (2024) offers a more nuanced view of this relationship. Using fMRI to examine patterns across the network during movie-watching and at rest, they observed both functional and structural connectivity patterns among intrahemispheric core regions (Baldassano et al., 2016) and found a strong alignment between the two; for example, the anterior PPA and RSC demonstrated both types of connectivity with the hippocampus (Kravitz, Saleem, et al., 2011; Watson & Andrews, 2024). There was also a strong alignment between anatomical position and structural connectivity; the anterior PPA showed stronger connections to ventral structures involved in memory and navigation, while the posterior PPA more strongly connected with dorsal areas processing

early visual features (Baldassano et al., 2016; Watson & Andrews, 2024). This bias suggests that the PPA’s anatomical position within the ventral and dorsal streams does shape its connectivity to some extent. Another key finding from this study were major white matter tracts along the network; notably, the ventrally located PPA connects with the inferior longitudinal fasciculus, which supports visual functions of the ventral stream (Herbet et al., 2018; Rokem et al., 2017). Additionally, the posterior PPA links with the vertical occipital fasciculus, which is thought to facilitate interaction between the ventral and dorsal pathways (Takemura et al., 2016). This study establishes robust anatomical connections between core visual areas, prompting further research into how anatomical placement affects the functionality of a region and its connection to the larger network.

We now reconvene back to the main question of this study: how do we define the PPA’s functionality? Past research has made clear that the PPA is sensitive to the spatial properties of a scene, hence the name “place area” (Epstein, 2008; Kravitz, Peng, et al., 2011; Park et al., 2011). However, we also understand that its unique medial position between two highly interactive pathways deems it more than just a “scene-selective” region (Kravitz, Peng, et al., 2011; Harel et al., 2013). Does the PPA’s structural connectivity with the ventral and dorsal streams translate into a broader functional role, serving as an “integration hub” for the merging of both object and scene information?

Past research on the PPA has primarily emphasized using static stimuli to assess differences between scene categories. For example, Epstein and Kanwisher (1998) presented static images to their participants to measure their brain responses in a controlled setting—as did Bar et al. (2008), Mullally & Maguire (2011), and Kravitz, Peng, et al. (2011). While this methodology is useful for studying distinct, fundamental connections within the visual network, it falls short when replicating real-world visual experiences that often require dynamic and complex neural connections. Previous studies using dynamic stimuli reinforce the idea that the viewing of dynamic environments amplifies the brain’s activity, providing a more accurate view of visual perception outside of controlled laboratory conditions (Pitcher et al., 2019; Küçük et al., 2024; Kröll et al., 2023). For instance, studies like Hasson et al. (2004) and Hanke et al. (2014) demonstrated that the viewing of continuous, dynamic scenes engage multiple brain regions differently, and more effectively, than isolated static images. Moreover, recent reviews explain how naturalistic stimuli better captures the interactions between cognitive processes during visual perception (Sonkusare et al., 2019; Nastase et al., 2021; Saarimäki, 2021; Küçük et al., 2022). To put this into perspective, we can imagine a person running through a forest compared to them standing idly in one position. The PPA, under the pressure of navigating through a rapidly changing environment, would have significantly different relationships with regions such as the RSC, in both amplitude and pattern of functional connectivity. To understand how the PPA functions and interacts with other regions in everyday visual processing, we must leverage the use of dynamic stimuli in the literature.

Using naturalistic stimuli from the StudyForrest dataset, where participants watched a two-hour movie in the fMRI scanner, Watson and Andrews (2024) produced a more comprehensive mapping of larger-scale brain connectivity. While this method enables us to understand the neural dynamics of naturalistic movie watching, it fails to consider an extra detail: movie segments are not all equal in how they activate the neural network. Movies contain a blend of both static and dynamic scenes and examining their patterns separately—not altogether—may help distinguish key relationships that form exclusively in dynamic environments.

In this study, I aimed to understand the role of the PPA in real-world scene perception. I hypothesize that the PPA serves as an integration hub for the neural network for scene perception; therefore, its interactions with other regions within the network is modulated by the perceptual demands of the visual scene. To test this hypothesis, I analyze the PPA’s activity and compare it under two distinct naturalistic conditions using the StudyForrest dataset: static scenes with limited movement versus navigational scenes with high dynamicity (Fig. 1). Static scenes allow the visual cortex to process spatial layouts without the added complexity of dynamic, moving elements, mimicking the controlled state of brain regions such as when examining a photo or a still environment. Due to the abundance of studies that have explored this exact phenomenon, the static stimuli are used as a baseline, or control, for understanding the brain’s function under the most basic, stable conditions. In contrast, navigational scenes, which follow a character’s movement, activate rapid processing of both spatial and object information (Thorpe et al., 1996; Fei-Fei et al., 2005; Rousselet et al., 2005). These scenes are expected to be essential to revealing how the PPA prioritizes connectivity between the ventral and dorsal streams under real-world environmental conditions. If navigational scenes challenge the PPA to rapidly integrate both spatial and object information (Bar & Aminoff, 2003; Bar, 2004; Rajimehr et al., 2008), this comparison will help us further define the PPA’s functionality and broader role within the visual network.

Methods

Dataset

I obtained data for 15 participants from an extension of the publicly available StudyForrest dataset (Hanke et al., 2014, 2016). The participants watched a two hour segment of the audio-visual movie “Forrest Gump” (<https://www.nature.com/articles/s41597-019-0303-3>) while undergoing 3T full-brain blood-oxygen-level dependent (BOLD) fMRI scanning as well as simultaneous 1000 Hz eye tracking. Functional data were collected from eight separate movie segments, each run approximately 15 minutes long with short breaks in between. Anatomical images for each participant were acquired separately from the 7T fMRI StudyForrest dataset (<https://www.nature.com/articles/sdata20143>).

Stimuli Breakdown

a) Static



b) Navigational

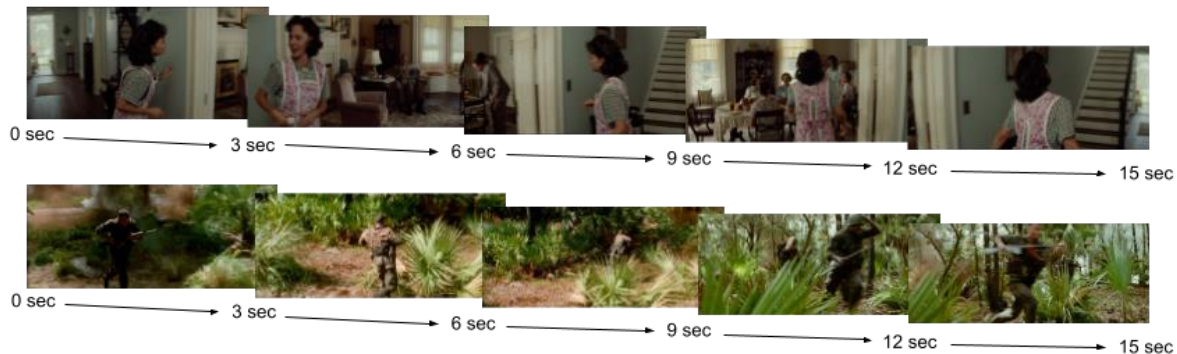


Fig. 1. Examples of static and navigational stimuli. Frames are pulled from 15-second duration scenes, broken up into 3-second intervals for visualization. (a) Static scenes included scenes with little to no movement of the camera and object. (b) Navigational scenes included scenes with the camera following the movement of a character.

To observe the functional connectivity patterns associated with static vs. navigational scenes, I selected 20 scenes from the film *Forrest Gump*, provided by the StudyForrest stimuli annotations. The scene categorization was conducted manually, given specific criteria (Fig. 1). Static scenes were defined as still shots with minimal camera and object movement, such as a character standing still in the dark or an empty meadow with no movement (Fig. 1a). In contrast, navigational scenes were selected based on dynamic camera movements following the path of a moving (e.g., walking or running) character (Fig. 1b). For example, both a soldier running through a forest and a character navigating their way through a house could be considered a navigational scene if the camera is directly accompanying their movement.

I chose scenes with a minimum duration of 14 seconds to allow for a sufficient neural response to be captured during fMRI scanning. This duration was chosen based on previous studies which have suggested a duration of 12-

15 seconds for the BOLD (Blood Oxygen Level Dependent) signal to stabilize and for a proper measurement of brain activity (Miezin et al., 2000; Buxton et al., 2004; Handwerker et al., 2012). By categorizing static and navigational scenes in this manner, I aimed to contrast low-motion, spatially limited conditions (static) with high-motion, spatially rich conditions (navigational), to explore the connectivity patterns that arise due to activation of spatial and navigational processing.

Preprocessing

Preprocessing of functional and structural data was performed using the CONN functional connectivity toolbox (Whitfield-Gabrieli & Nieto-Castanon, 2012) and Statistical Parametric Mapping (SPM12) software (Wellcome Trust Centre for Neuroimaging; <http://www.fil.ion.ucl.ac.uk/spm>). CONN's default preprocessing pipeline was used for volume-based analyses to realign and unwarp all functional images, as well as identify head motion outliers. Expressions to follow are directly from the CONN toolbox: Slice-timing correction was applied using the "interleaved bottom-up" option in SPM12. The functional and structural images were then segmented and normalized to the standard Montreal Neurological Institute (MNI) space using the default tissue probability maps (TPM) found in SPM12. The target resolution for the functional images were set to 3 mm to optimize file size and preprocessing time. The normalized functional data was smoothed using spatial convolution with a Gaussian kernel of 6mm full width half maximum (FWHM). Following the preprocessing steps and quality assurance (QA) checks, denoising was operated using the CONN toolbox, applying a band-pass filter with a frequency range of 0.008 to 0.09 Hz to the smoothed data. Lastly, CheckReg in the SPM12 interface was used to ensure that the resulting images were in alignment with the T1-weighted anatomical MRI scans of each participant.

Regions of Interest

The regions of interest (ROI) used in this study were the PPA, OPA, RSC, and LOC. Both hemispheres were considered separately, resulting in a total of eight ROIs for this study (Fig. 2). To determine reliable MNI coordinates (x, y, z) for each region, the peak coordinates from various neuroimaging studies were gathered, and the x-, y-, and z-coordinates were averaged separately. The resulting peak MNI coordinates for each region were rounded to the nearest whole number for simplicity and plotted using the WFU_PickAtlas tool in the SPM12 interface. Spherical ROIs were then created with a radius of 7.5 mm for each region; the appropriate radius was determined by taking the average across all standard deviation values calculated from the mean coordinates of various neuroimaging studies. To promote simplicity, this average was collapsed across all ROIs and the same radius was applied for each spherical region.

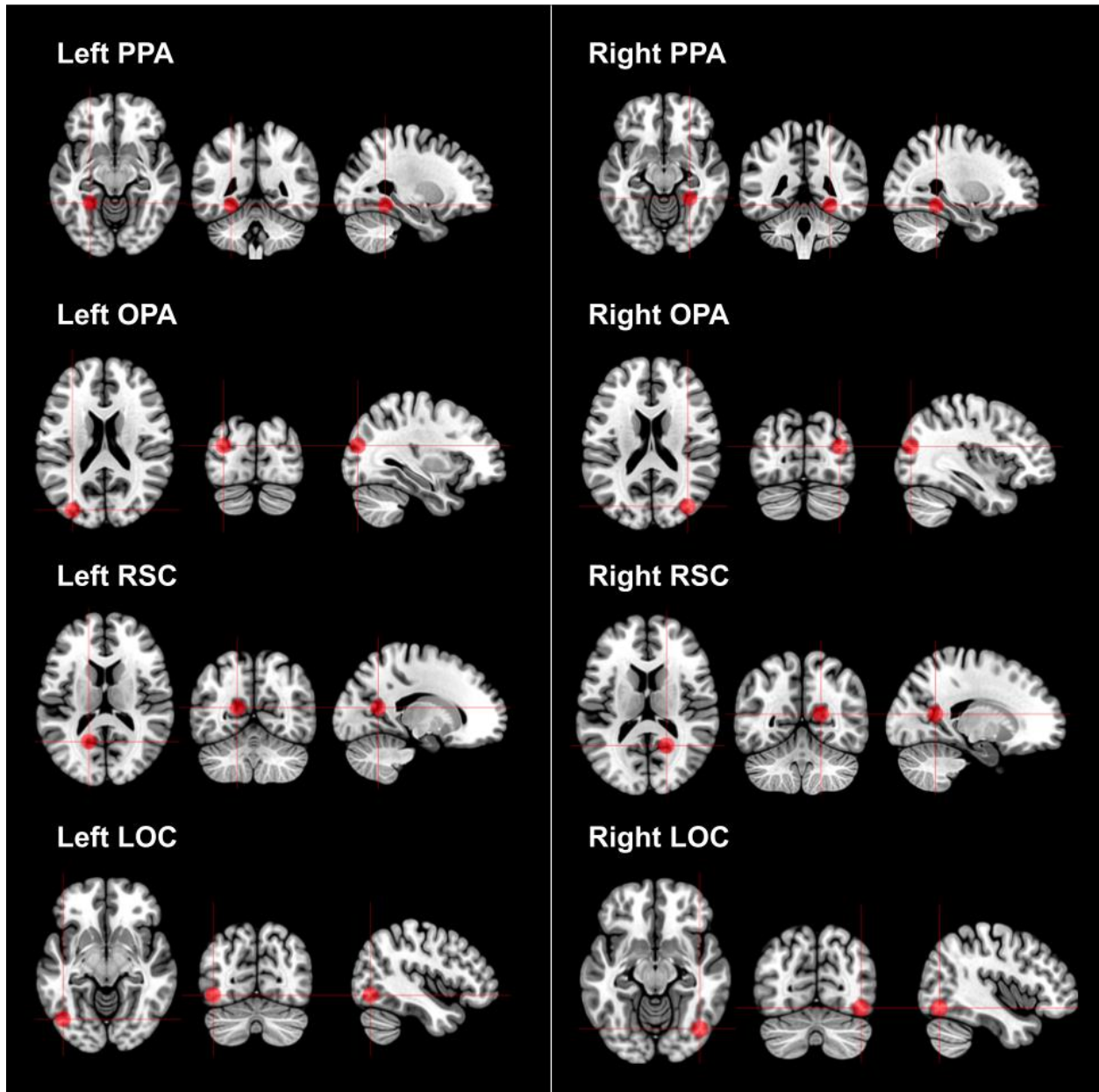


Fig. 2. Regions of Interest (ROIs) displayed on a standard MNI template (mni152) using MRIcroGL. MNI coordinates for each region were averaged from the peak coordinates from various neuroimaging studies (Table 1).

Functional Connectivity

I began my functional connectivity analysis by extracting the average BOLD signals from each ROI for each subject across the entire movie. Using the timestamps of the stimuli annotations, I then categorized the BOLD signals by scene and saved them separately according to the scene category (static or navigational).

With the categorized BOLD signals, I calculated the Pearson correlation coefficients between each pair of ROIs, creating 8×8 correlation matrices for each subject and each scene category. These correlations represent the functional connectivity between brain regions of an individual subject. I then averaged the correlations across all subjects to create final correlation matrices for both static and navigational scenes, aiding in group-level analysis of functional connectivity patterns. To determine the significance of the differences in correlation coefficients between ROI-pairs across static and navigational scenes, I conducted paired t-tests and displayed the t-values and p-values on

a colored heatmap. A threshold of 0.05 for the p-value was used in correspondence to the convention (Fisher, 1925). This parametric test assessed whether the difference in functional connectivity measures between static and navigational scenes was statistically significant.

Graph Metrics

I further explored connectivity patterns using graph-theory metrics, which offer a quantitative framework for analyzing brain connectivity (Watts & Strogatz, 1998). The final averaged correlation matrices for each scene category were used to construct network graphs, where nodes represent ROIs and edges represent the correlation coefficients, or the strength of functional connectivity, between them (Sporns & Zwi, 2004; Bullmore & Sporns, 2009). To account for MATLAB's default interpretation of high edge weights as longer path lengths, the correlation coefficients were inverted before any calculations. A threshold of 0.1 was then applied to filter out spurious connections, ensuring that the graph is reliable and not overly dense. Thresholding with values such as 0.1 is a common practice used in the literature to reduce noise (Bassett & Bullmore, 2009). I used the resulting graphs to study five local graph measures: degree, clustering coefficient, characteristic path length (CPL), modularity, and betweenness centrality (Abazid et al., 2021). Among these metrics, I found that the betweenness centrality, modularity, and CPL were particularly unique in distinguishing between static and navigational scenes.

Betweenness centrality measures the number of shortest paths that pass through a particular node (ROI) (Wasserman & Faust, 1994). ROIs with a greater number of shortest paths passing through serve as critical hubs for information flow and are of greater importance within the network (Freeman, 1977). I compared the betweenness centrality values of each ROI to explore whether certain regions, such as the PPA, function as a hub for information processing between the selected ROIs.

The modularity score (q) measures the presence of a community structure within the network (Newman, 2004, 2006, 2011; Danon et al., 2005; Blondel et al., 2008; Ruan & Zhang, 2008; Meunier et al., 2009). High modularity indicates that the network has strong and dense connections within modules, or clusters of regions, but weaker and sparse connections between separate modules (Newman, 2004, 2006, 2011). In this study, modularity is used to compare brain network organization between static and navigational scenes: does one scene type result in a stronger inter-network integration than the other?

The characteristic path length (CPL) represents the average shortest path between all pairs of nodes (ROIs), essentially providing a measure of the network's overall efficiency in transmitting information (Burgstaller et al., 2019). Thus, a shorter CPL value implies more efficient communication, whereas a larger value indicates longer transmission routes (Lovejoy & Loch, 2003; Pamplona et al., 2015). In this study, I computed the CPL values for static and navigational scenes, predicting that navigational scenes, requiring bidirectional scene and object processing, would result in a shorter CPL for greater efficiency.

Upon gathering the graph metrics, a meaningful difference was observed in the CPL between the two scene types. I conducted a post-hoc nonparametric permutation test to assess this difference. The observed difference was calculated by subtracting the average CPL from static scenes from that of navigational scenes. I then ran all possible permutations with the 15 subjects ($32,767 = 2^{15} - 1$) to construct the null distribution. That is, within each permutation, I randomly shuffled the scene category labels, generated the average permuted matrices, and calculated the permuted CPL difference. The distribution of CPL differences generated from 32,767 cases accounted for the null hypothesis, under which no differences in the CPL values exist between the two scene categories. The p-value was computed by comparing this null distribution against the observed difference.

Results

In analyzing the experimental results, I was primarily interested in how the PPA interacts with other regions during both static and navigational scenes. Unexpectedly, however, the study also yielded several notable hemispheric differences in the connectivity patterns.

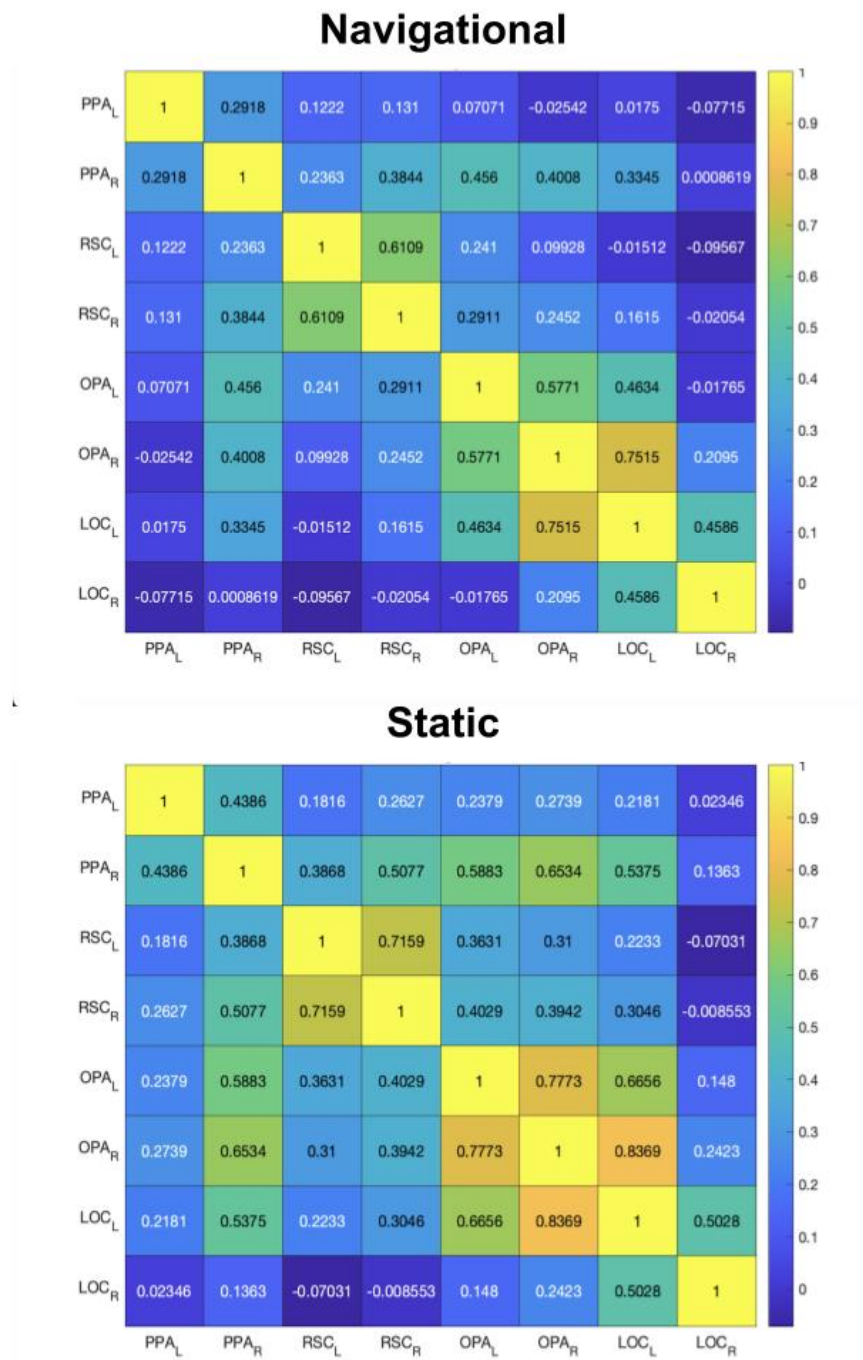


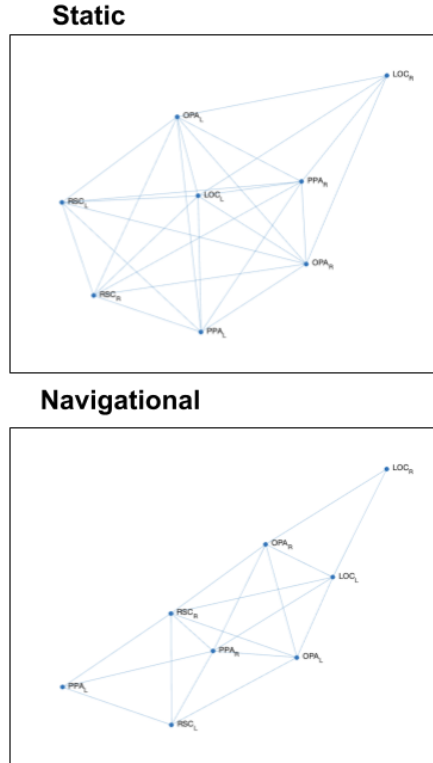
Fig. 3. Final correlation matrices for static and navigational scenes, averaged across all 15 subjects. Axes are labeled with all 8 ROIs, each box containing the correlation coefficient between two ROIs. Overall, correlations for static scenes are markedly higher than those for navigational scenes.

Unique Hemispheric Differences in the PPA

Our analysis revealed significant differences in the functional connectivity between the right and left hemispheric PPA, with the right PPA (PPA_R) demonstrating consistently stronger connectivity with other regions of interest

(ROIs) than the left PPA (PPA_L) regardless of the scene type. Although this pattern was not expected to be a priority, a paired t-test was performed to quantify the differences in the correlation coefficients. The final correlation matrices for both static and navigational scenes displayed higher correlation coefficients for all relationships with PPA_R than PPA_L, with p-values indicating the differences are statistically significant ($p = 0.0002$ for static; $p = 0.0000$ for navigational) (Fig. 3). Moreover, PPA_R had a higher betweenness centrality (4 in static, 6 in navigational) compared to PPA_L, which had a betweenness centrality of 0 in both conditions (Fig. 4b). The high betweenness centrality suggests that information flows frequently through PPA_R to reach other parts of the network, which is crucial for bi-directional communication (Freeman, 1979; Pacheco et al., 2019).

(a) Connectivity Graphs



(b) Betweenness Centrality

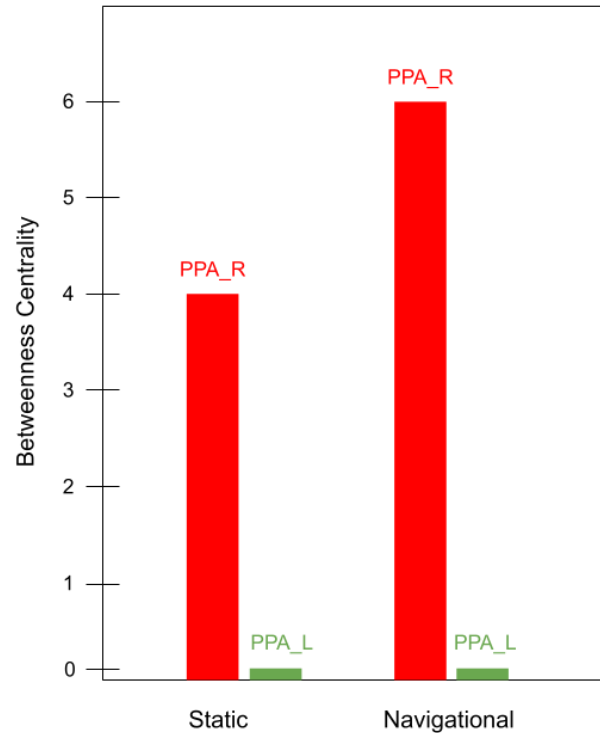


Fig. 4. (a) Connectivity graphs for static and navigational scenes mapping out the connections between ROIs. Each node represents a specific ROI (labeled accordingly) and lines represent the connectivity between the two regions. Shorter path lengths (edges) represent stronger correlations, and longer path lengths represent weaker correlations. (b) A bar graph depicting the betweenness centrality values for the right PPA (PPA_R) and the left PPA (PPA_L) in both static and navigational scenes.

These results strongly suggest that PPA_R plays a significant role in coordinating information flow between the ventral and dorsal pathways (Pacheco et al., 2019). However, it also indicates that PPA_L is uniquely different from PPA_R, playing a lesser role in this integration. Accounting for the hemispheric differences, I changed the course of the analysis to focus on PPA_R's integrative role between the ventral and dorsal stream, with the goal of exploring its alignment with the second hypothesis. Correspondingly, I decided to exclude PPA_L from further analysis of the static and navigational scene categories.

The Right PPA Shows Higher Connectivity During Static Scenes Than Navigational

After observing PPA_R's overall strong connectivity with other regions, I compared its connectivity patterns between static and navigational scenes. A nonparametric permutation test revealed no significant difference in its betweenness centrality between static ($bc = 4$) and navigational ($bc = 6$) ($p = 0.6382$; above the 0.05 threshold) (Fig.

4b). However, after running a paired t-test on both correlation matrices, I found that PPA_R's connectivity was significantly stronger with regions across both the ventral and dorsal pathway (LOC_L, OPA_R, OPA_L, RSC_L, PPA_L) during static scenes than navigational ($p < 0.05$) (Fig. 5). This may serve as evidence of heightened communication between the two pathways during static scenes compared to navigational scenes.

Correlation Differences: Static - Navigational

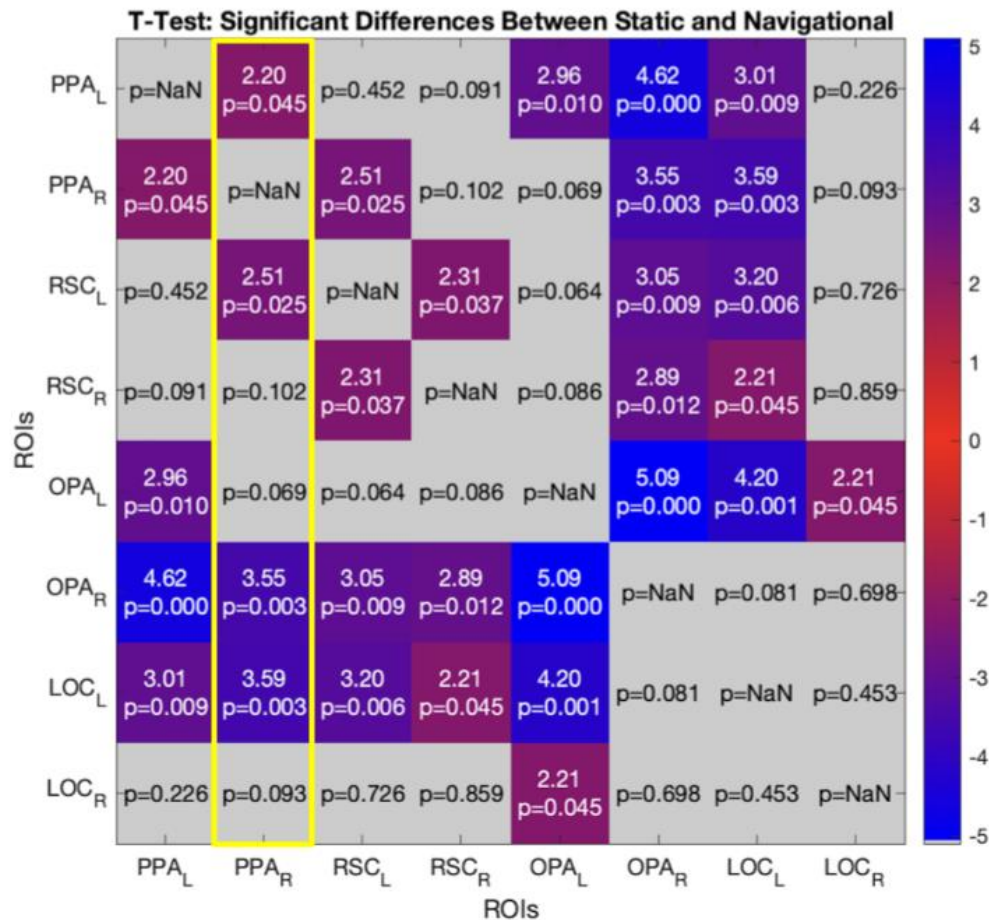


Fig. 5. Differences in correlations between static and navigational scenes, as determined by a series of pairwise t-tests. A positive difference indicates greater correlation during static scenes, and vice versa. If $p < 0.05$, the difference is considered statistically significant. Notably, PPA_R has significantly higher correlations with LOC_L, OPA_R, OPA_L, RSC_L, and PPA_L during static scenes than navigational scenes.

Static Scenes Reflect a More Globally Integrated Network Than Navigational Scenes

I then ran a post hoc analysis to test any differences in entire-network communication between the two scene types. Again, using a nonparametric permutation test, I found that the modularity score was significantly higher for navigational scenes ($q = 0.3253$) compared to static scenes ($q = 0.1960$), with a p-value of 0.0013 ($p < 0.05$) (Fig. 6a). Similarly, characteristic path length was significantly greater during navigational scenes ($CPL = 3.919$) compared to static scenes ($CPL = 2.701$), with a p-value of 0.0009 ($p < 0.05$) (Fig. 6b).

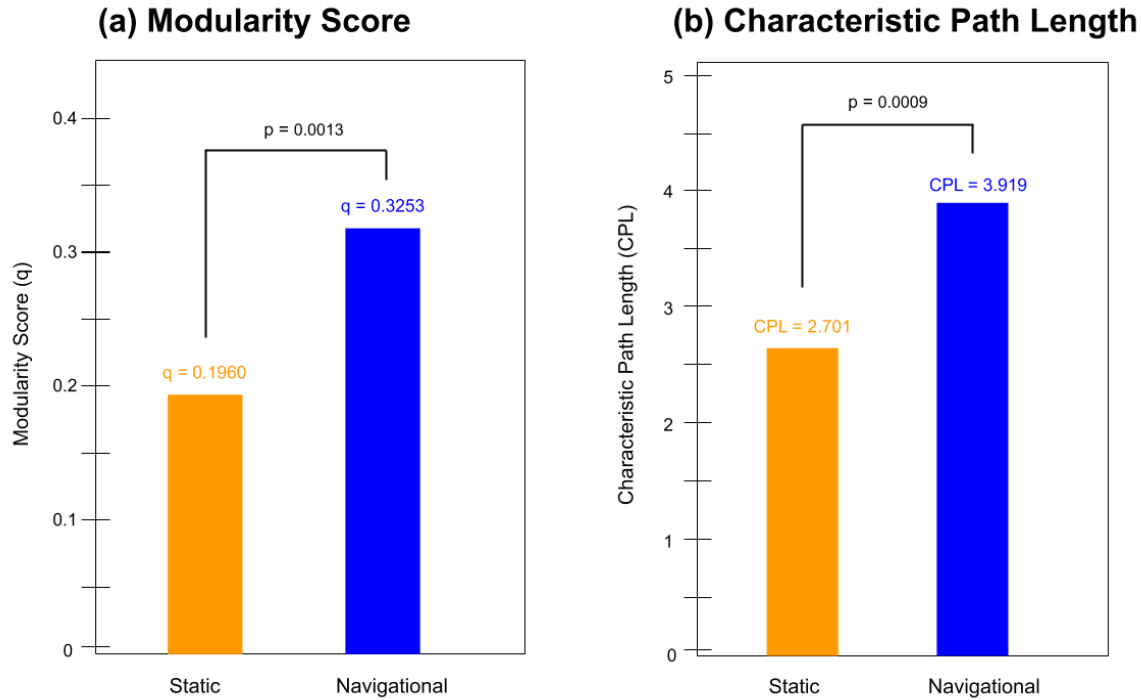


Fig. 6. (a) Comparison of modularity scores (q) between static and navigational scenes. Modularity was significantly higher for navigational scenes compared to static scenes ($p = 0.0013$), indicating a more modular network during navigational scenes. (b) Comparison of characteristic path lengths (CPL) between static and navigational scenes. CPL was significantly higher for navigational scenes compared to static scenes ($p = 0.0009$), suggesting longer communication routes during navigational scenes.

These findings unexpectedly contrast with my original prediction that navigational scenes, due to their higher cognitive demands, should have a lower modularity score and characteristic path length—both measures combined suggesting a highly interconnected network with shorter, more efficient communication (Pamplona et al., 2015; Pacheco et al., 2019). An interpretation of the results contrary to my expectation is discussed in the Discussion.

Discussion

This study investigated the function of the PPA in relation to its unique anatomical position in between the ventral and dorsal pathways; does its connections with other regions reflect a larger role as an “integration hub”? I addressed this question by analyzing functional connectivity patterns among the network of brain regions including the PPA under one condition when participants watched a static scene from a movie versus another condition when participants watched a dynamic navigational scene from the movie. The results of the experiment unexpectedly revealed significant hemispheric differences between the right and left PPA, impelling us to consider the two hemispheres separately when answering the study’s objective. Importantly, the results also examined stark differences between responses to static and navigational scenes, which offer a new perspective on entire network integration.

The Right PPA Functions as an “Integration Hub”

The right PPA showed much stronger connectivity with other brain regions than the left PPA, as demonstrated by its high betweenness centrality across both scene types (Fig. 4b). As depicted on both the static and navigational graphs, the right PPA lies in the middle of the network and connects the object-selective region LOC with scene-selective regions RSC and OPA (Fig. 4a). The high betweenness centrality value, significantly larger than that of

other ROIs, suggests that the right PPA is a crucial “connector” in the network, bridging information flow between the ventral and dorsal pathways (Pacheco et al., 2019). One interpretation of this finding is that the right PPA largely serves as an “integration hub” rather than just a region strictly specialized for scene processing.

To understand this in everyday terms, we can think of the right PPA like a central train station that transports passengers (object and scene information) to multiple stops (object- and scene-selective regions). The region catalyzes information flow between the two specialized modules for greater efficiency. This further supports the PPA’s anatomical position between the two streams and explains why object and scene processing were found to be highly interactive in previous related studies (Troiani et al., 2014; Brandman & Peelen, 2017, 2023).

The left PPA, on the other hand, reflected a more minor role in this interaction, displaying low connectivity across the network and zero betweenness centrality regardless of the scene type (Fig. 4b). In terms of the previous analogy, we can view the left PPA as a smaller, more local train station that only serves passengers from one block to the next. However, despite having low connectivity across the board, the left PPA surprisingly reflected a similar pattern as the right PPA, in that it consistently showed relatively higher connectivity with other regions during static scenes than navigational. Although this result does not add to our understanding of the left PPA’s functionality, it does reinforce the distinction discussed earlier: static scenes evoke higher connectivity across the entire network, while navigational scenes consistently display the opposite (Fig. 3).

Overall, the hemispheric differences between the left and right PPA challenge the idea that both sides of the PPA contribute equally to scene perception. Instead, the results suggest that the right PPA plays a much more critical role in integrating information from different parts of the brain. Future research should aim to understand why these hemispheric differences occur.

Static and Navigational Scenes Evoke Distinct Responses from the Neural Network

The right PPA displayed statistically stronger connectivity with other regions during static scenes than navigational scenes. This was unexpected, as I hypothesized that navigating a dynamic environment would require more integration between the ventral and dorsal pathways to make sense of the scene’s complexity.

However, a possible explanation is that a static environment allows the brain to process the entire scene at once, requiring increased connectivity between both pathways to form a complete representation. In everyday life, this might be similar to standing still and scanning a room—with ample time and capability, the brain focuses on integrating all the visual elements into a coherent picture. In contrast, navigational scenes are highly dynamic and require adaptation to immediate changes, such as when navigating through a winding trail down a forest. The short processing time and rapidly changing perception make it impossible for all modules to work together in integrating each visual element into the picture. Instead, each specialized module must contribute their part independently of the others to achieve maximum efficiency, hence the weaker functional connectivity. The increase in localized processing could explain why functional connectivity with the right PPA generally decreases during navigational scenes, even though the region remains the central “integration hub.”

Differences between static and navigational scenes become even more apparent when looking at the entire network. Static scenes had a shorter characteristic path length and modularity score, exhibiting a more globally integrated network where different brain regions communicate easily and equally with each other (Fig. 6). In real life, this could be like working in a small, well-connected team, where information flows freely between all members and no one relationship is stronger or weaker than the rest.

On the other hand, navigational scenes showed higher modularity and a longer characteristic path length, indicating that the brain is more fragmented with regions working independently and forming specialized groups for processing different aspects of the scene. This specialized processing, where regions are forced to prioritize certain connections over others (Hilgetag et al., 2000; Bullmore & Sporns, 2009; Sporns & Betzel, 2016), is essential for the brain to compartmentalize different tasks (i.e. identifying landmarks and avoiding obstacles) without needing to integrate all the information at once. Therefore, navigational scenes require a modular network in comparison to static scenes, which reflects more global connectivity.

The distinction between static and navigational scenes provides significant insight into the brain’s adaptability to different environmental demands. Particularly, the strong connectivity possible during static scenes indicate that the brain prioritizes comprehensive processing of its surrounding environment when time and perceptual stability allow. We can also interpret the PPA’s functionality to be context-dependent; it does not serve as an integration hub over a scene-selective region, or vice versa. Rather, its unique anatomical position allows it to prioritize connections between the ventral and dorsal pathways appropriately, depending on the cognitive demands of the visual environment.

Conclusion

In conclusion, this study supports the idea that the PPA functions as more than a strictly scene-selective region; it serves as a crucial connector between the ventral and dorsal pathway, allowing efficient neural communication between them for one complete understanding of the visual environment. The right hemispheric PPA seems to play a larger role in this integration of information compared to the left PPA. The viewing of static stimuli allows stronger functional connectivity across the network, exhibiting global integration of regions. In contrast, navigational stimuli require a modular network organization, where regions prioritize only a few connections to maximize the speed and efficiency of dynamic visual processing.

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